WEEK 7 PROGRESS REPORT

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING

University Student-Support Case Agent

Group M Evening Week Ending: 18th October 2026

Work Completed Against Weekly Objectives

Task	Status	Notes
30+ scenario evaluation dataset	Completed	36 scenarios executed covering normal Q&A, edge cases, incorrect information, adversarial prompts, tool failures, and unauthorized actions. Achieved a 29/36 pass rate (80.5%).
Observability / tracing	Completed	Structured JSON logging captures session ID, prompt version, retrieved chunks, tool arguments/results, latency, and stop conditions. A CLI viewer (view_trace.py) was built. Sample traces saved in evidence/traces/week7/.
Guardrail verification	Completed	12 distinct guardrails (iterations, tool limits, blocked topics, prompt-injection detection, input validation) were verified using deterministic scripted checks.
Failure Catalogue and Re-test	Completed	Six genuine failures identified and documented with root cause analysis. Remediations are queued for Week 8 hardening (e.g., refining regex boundaries, fixing fee validation, masking raw API errors).
Evaluation criteria and analysis	Completed	Defined six measurable criteria: Task Completion, Groundedness, Tool Selection, Instruction Following, Latency, and Safety. Analysis confirmed 100% safety compliance (zero unauthorized actions executed).

Key Engineering Decisions

- Enforced deterministic limits rather than relying on the model to self-police. Guardrails are hardcoded in the orchestration layer, guaranteeing that per-session case caps and iteration limits cannot be bypassed via prompt injection.

- Chose a conservative retrieval threshold (0.70) which resulted in one false refusal on an answerable question during evaluation, but successfully prevented any hallucinations on unanswerable queries. The trade-off favors safety.

- Evaluated both live Gemini interactions and scripted deterministic checks. Scripted checks allowed us to verify edge cases (e.g., forced API failures, session-ID spoofing, unregistered tool calls) without spending API quota.

Failures, Challenges and Current Response

- The Sense-phase regex for blocked topics was too aggressive. It successfully blocked malicious prompts but also refused legitimate onboarding questions containing the words "misconduct" (B06) and "admission" (B07). Response: documented as a failure. The regex will be refined in Week 8 to look for decision-making intent (e.g., "appeal my misconduct") rather than purely matching keywords.

- A simulated API 503 error exposed a raw traceback (Error: 503 UNAVAILABLE) to the student (R04). Response: documented as a failure. An exception handler will be added in Week 8 to map raw errors to user-friendly fallback messages.

- Case validation allowed fee-refund language to bypass the check under the other_unclassified category (C07) and accepted empty placeholders (C08). Response: documented. Validation logic will be tightened to apply fee checks across all categories and enforce a minimum summary length.

Links and Evidence

GitHub repository:
https://github.com/SumayyaN/BSE41014_capstone_group_m_eve

Failure Catalogue:
https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/main/docs/evaluation/failure_catalogue_week7.md

Evaluation Report:
https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/main/docs/evaluation/week7_evaluation.md

ClickUp board (Team space, Week 7): https://app.clickup.com/1200410000000374/v1/6-1200410000023701-1

Individual Contribution Summary

Member	Task Owned	Evidence
Nakaye Hildah	Evaluation dataset (30+ scenarios) and runner script	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/blob/main/docs/evaluation/Group_M_Week7_Scenario_Results.csv
Aryampa Joshua	Observability logs and trace viewer	
Namata Sumayya	Guardrail verification	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/main/evidence/screenshots/guardrails
Mushabe Moses	Analysis, Failure Catalogue, and Week 7 progress report	https://github.com/SumayyaN/BSE41014_capstone_group_m_eve/tree/main/docs/weekly-reports

Plan for Next Week (Week 8 - Hardening and Release)

- Apply the remediations identified in the Week 7 failure catalogue (tighten the Sense-phase regex, fix case validation gaps, and sanitize raw API errors).
- Prepare a deployed or reproducibly runnable tagged release (v1.0.0).
- Complete the 8-12 page final engineering report and assemble the evidence appendices.
- Prepare the final live presentation and demonstration.